

Driven systems:

$$m\ddot{x} + \gamma\dot{x} + kx = f(t)$$

li) Homogeneous solⁿ: $x_H(t)$

ki) Particular solⁿ: $x_p(t)$

→ can be done by guessing

lci) Solⁿ $x = x_H(t) + x_p(t)$

$$x(0) \equiv x_0$$

$$\dot{x}(0) \equiv v_0$$

ex: $m\ddot{x}_s + \gamma_s \dot{x}_s + kx_s = f \sin(\omega_s t_s)$

$$\hookrightarrow \ddot{x} + 2\gamma\dot{x} + x = \sin(\omega t)$$

• Particular solⁿ: $x_p(t) = c \sin(\omega t) + d \cos(\omega t)$

$$\ddot{x} = -\omega^2 x$$

$$\dot{x} = \omega (c \cos(\omega t) - d \sin(\omega t))$$

$$(-\omega^2 + 1)(c \sin(\omega t) + d \cos(\omega t)) + 2\gamma\omega (c \cos(\omega t) - d \sin(\omega t)) = \sin(\omega t)$$

$$0 = \sin(\omega t) (-\omega^2 c + c - 1 - 2\gamma\omega d) + \cos(\omega t) (-\omega^2 d + d + 2\gamma\omega c)$$

$$\begin{pmatrix} 1 - \omega^2 & -2\gamma\omega \\ 2\gamma\omega & 1 - \omega^2 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

inverse: $\frac{1}{(1-\omega^2)^2 + 4\gamma^2\omega^2} \begin{pmatrix} 1-\omega^2 & 2\gamma\omega \\ -2\gamma\omega & 1-\omega^2 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$$c = \frac{1-\omega^2}{(1-\omega^2)^2 + 4\gamma^2\omega^2}$$

$$d = \frac{-2\gamma\omega}{(1-\omega^2)^2 + 4\gamma^2\omega^2}$$

$$x(t) = c \sin(\omega t) + d \cos(\omega t) + x_H$$

ex 2: $\ddot{x} + 2\gamma\dot{x} + x = f(t)$

$$f(t) = \begin{cases} 1 & , t \in [0, \tau] \\ 0 & \text{else} \end{cases}$$

$$\gamma < 1, \quad x(0) = 0 \quad \dot{x}(0) = 0$$

split ODE:

$$\left(\frac{d}{dt} + \gamma - i\sqrt{1-\gamma^2} \right) \underbrace{\left(\frac{d}{dt} + \gamma + i\sqrt{1-\gamma^2} \right)}_{\omega} x = f(t)$$

$$\frac{d\omega}{dt} + (\gamma - i\sqrt{1-\gamma^2})\omega = f(t)$$

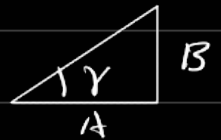
$$\text{or } \ddot{x} + 2\gamma\dot{x} + x = f$$

$$x_p(t) = 1$$

$$x(t) = x_p(t) + x_H(t)$$

$$\bullet A \cos(\Omega t) + B \sin(\Omega t)$$

$$\tan(\gamma) \equiv \frac{B}{A}$$



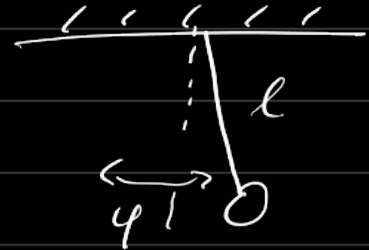
$$A \left(\cos(\Omega t) + \frac{\sin(\gamma) \sin(\Omega t)}{\cos(\gamma)} \right) = A \frac{\cos(\Omega t - \gamma)}{\cos(\gamma)}$$

$$\cos(\gamma) = \frac{A}{\sqrt{A^2 + B^2}} \Rightarrow \frac{A}{\cos(\gamma)} = \sqrt{A^2 + B^2} \equiv C$$

$$= C \cos(\Omega t - \gamma)$$

$$m \ddot{\vec{r}} = \vec{F} + m (\vec{\omega} \times (\vec{\omega} \times \vec{r}_{\Sigma}))$$

$$m \ddot{\vec{r}} = -mg \hat{e}_z + m (\vec{\omega}$$



$$\vec{\omega} = \vec{v} \times \vec{r}$$

